

Extra exercise

The covariance matrix of (X, Y) is $\Sigma = \begin{bmatrix} \sigma_x^2 & \text{cov}(X, Y) \\ \text{cov}(X, Y) & \sigma_y^2 \end{bmatrix} = \begin{bmatrix} \sigma_x^2 & \rho \sigma_x \sigma_y \\ \rho \sigma_x \sigma_y & \sigma_y^2 \end{bmatrix}$

The joint distribution is (see chap.1)

of the multivariate normal (assuming $\mu_x = \mu_y = 0$):

$$f(x, y) = \frac{1}{2\pi \sqrt{|\Sigma|}} \exp \left[-\frac{1}{2} (x, y) \Sigma^{-1} \begin{pmatrix} x \\ y \end{pmatrix} \right]$$

calculating Σ^{-1} , we find $\Sigma^{-1} = \frac{1}{1-\rho^2} \begin{bmatrix} \frac{1}{\sigma_x^2} & -\frac{\rho}{\sigma_x \sigma_y} \\ -\frac{\rho}{\sigma_x \sigma_y} & \frac{1}{\sigma_y^2} \end{bmatrix}$

$$|\Sigma| = \begin{vmatrix} \sigma_x^2 & \rho \sigma_x \sigma_y \\ \rho \sigma_x \sigma_y & \sigma_y^2 \end{vmatrix} = \sigma_x^2 \sigma_y^2 (1-\rho^2)$$

$$(x \ y) \Sigma^{-1} \begin{pmatrix} x \\ y \end{pmatrix} = (x \ y) \begin{pmatrix} \frac{x}{\sigma_x^2} - \frac{\rho y}{\sigma_x \sigma_y} \\ -\frac{\rho x}{\sigma_x \sigma_y} + \frac{y}{\sigma_y^2} \end{pmatrix} \times \frac{1}{1-\rho^2} = \frac{1}{1-\rho^2} \left(\frac{x^2}{\sigma_x^2} - \frac{2\rho xy}{\sigma_x \sigma_y} + \frac{y^2}{\sigma_y^2} \right)$$

So $f(x, y) = \frac{1}{2\pi \sigma_x \sigma_y \sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \left(\frac{x^2}{\sigma_x^2} - \frac{2\rho xy}{\sigma_x \sigma_y} + \frac{y^2}{\sigma_y^2} \right) \right]$

- The marginal distribution of X can be obtained by integrating $f(x, y)$ with respect to y : $f_x(x) = \int_{-\infty}^{\infty} f(x, y) dy$.

and this yields a Gaussian distribution - eg. $f_x(x) = \frac{1}{\sqrt{2\pi} \sigma_x} \exp \left(-\frac{x^2}{2\sigma_x^2} \right)$.

- From Chap.1 (Bayes law): $f(y|x=x_0) = \frac{f(x_0, y)}{f_x(x_0)}$

We obtain:

$$f(y=y|X=x_0) = \frac{1}{\sqrt{2\pi} \sigma_y \sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \left(\frac{x_0^2}{\sigma_x^2} - \frac{2\rho x_0 y}{\sigma_x \sigma_y} + \frac{y^2}{\sigma_y^2} \right) + \frac{1}{2} \frac{x_0^2}{\sigma_x^2} \right]$$

$$= \frac{1}{\sqrt{2\pi} \sigma_y \sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \left(\frac{x_0^2}{\sigma_x^2} - (1-\rho^2) \frac{x_0^2}{\sigma_x^2} - \frac{2\rho x_0 y}{\sigma_x \sigma_y} + \frac{y^2}{\sigma_y^2} \right) \right]$$

$$= \frac{1}{\sqrt{2\pi} \sigma_y \sqrt{1-\rho^2}} \exp \left[-\frac{1}{2} \frac{1}{\sigma_y^2 (1-\rho^2)} \left(y - \rho \frac{\sigma_y}{\sigma_x} x_0 \right)^2 \right]$$

$$\Rightarrow Y|X=x_0 = \mathcal{N} \left(x_0 \rho \frac{\sigma_y}{\sigma_x}, \sigma_y^2 (1-\rho^2) \right); \text{ i.e. } E(Y|X=x_0) = \rho \frac{\sigma_y}{\sigma_x} x_0 \propto x_0; \text{ var}(Y|X=x_0) = \sigma_y^2 (1-\rho^2).$$